AI Capstone
Homework 3: A Robot Navigation Framework
Announce: 2026/4/13, Due date: 2026/4/27

Introduction

In this assignment, you will use a semantic map of apartment_0 to navigate a robot to a specified destination (e.g., locating a target object). You will implement point-to-point navigation using the RRT algorithm.

(Only the first floor of apartment_0 will be used.)

There are mainly three parts:

Part 1. 2D semantic map construction
Use the provided semantic point cloud to get a 2D semantic map for navigation.

Part 2. RRT algorithm implementation
We use the RRT algorithm to find a navigable path from starting point A to goal point B (a specific item).

Part 3. Robot navigation
An agent can navigate automatically by following the path calculated by RRT to find specific items.

Implementation

You can either complete the TODOs in the provided template code or implement your own version as long as you follow the below guides.

Example Final Result (Target object: cooktop)

Part 1: 2D semantic map construction

The following describes the steps for generating a 2D semantic map.

1. We provide a 3D semantic map of the first floor of apartment_0 under

hw3/semantic_3d_pointcloud. Remove ceiling and floor points of the point cloud and do any extra processing you need.

2. Save coordinates and colors of the points (the color should be the same as color map for 101 categories)
3. Use any libraries (e.g. matplotlib) to plot a scatter graph (x-coordinate, z-coordinate in the point clouds) with the points you saved

Notes for Step 3:

- **The scale relationship** between coordinates in **point.npy** and apartment_0 is **apartment_0 = points array * 10000. / 255.**

- There are two versions of the color array. The value of RGB in **color01.npy** is in the range [0, 1] while **color0255.npy** is in the range [0, 255]. If you have problems using **color01.npy** to find labels because of floating-point errors, you can use **color0255.npy.**

Note: You need to find the relationship between the map (pixel) coordinate and the Habitat coordinates. It's necessary for **Part 3**. **You can add extra points to calculate the scale or create different maps to help you**.

Example map:

[After removing ceiling]

[After Removing ceiling and floor]

Part 2: RRT Algorithm

You will implement the RRT algorithm, which calculates a navigable path from the **2D map we created from part 1**. For more details on the algorithm, please check details on this link, RRT [1] , or the slide in the class.

You need to search for these target categories, including **rack, cushion, sofa, stair, and cooktop. You can check the color map for 101 categories for the corresponding colors.**

Input:

1. **A target category you want to search**
   Just input a string (e.g:, **rack**)

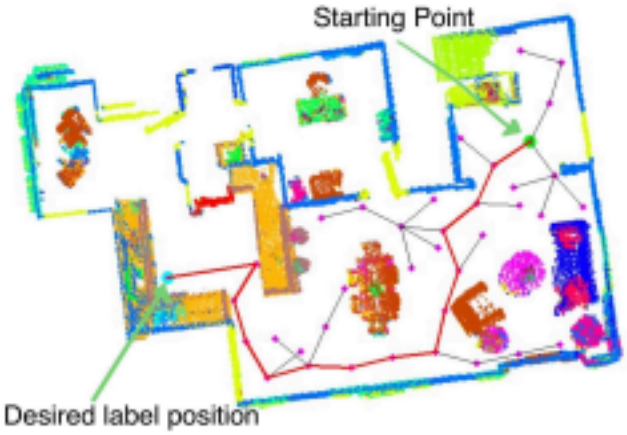
2. **Choose a starting point to navigate by clicking on the map (on the visualized GUI)**

Output:

1. **A map with a path from a starting point to a target point.** The target point is a point **in front of the target item**. You can use any method to decide the location of the point.

2. **Waypoints on this route**
   The xy-coordinate calculated by RRT is in the pixel coordinate. To navigate in Habitat, you need to convert them into the coordinate of **the first floor apartment_0.**

Example of a RRT output:



Part 3: Robot Navigation

1. Based on the path you get in Part 2, let the agent move using these three commands: "move_forward", “turn_left”, and “turn_right”.
   - You can modify it from hw0/load.py to do navigation or use hw3/navigator.py

- We can define the amount of how much the agent should move and turn for each step by changing the following config

```
# Define action space
agent_cfg.action_space = {
    MOVE_FORWARD: habitat_sim.agent.ActionSpec(
        MOVE_FORWARD, habitat_sim.agent.ActuationSpec(amount=MOVE_AMOUNT)
    ),
    TURN_LEFT: habitat_sim.agent.ActionSpec(
        TURN_LEFT, habitat_sim.agent.ActuationSpec(amount=TURN_AMOUNT)
    ),
    TURN_RIGHT: habitat_sim.agent.ActionSpec(
        TURN_RIGHT, habitat_sim.agent.ActuationSpec(amount=TURN_AMOUNT)
    ),
}
```

2. Please highlight the target with a **transparent mask** while navigating so that we can clearly visualize the target category.
   - Ref: [tutorial_adding_images]

Examples of a transparent mask

Grading

Online Demo 30%

We will specify a target label among: **rack, cushion, sofa, stair, and cooktop** and a starting point. You should show us the corresponding navigation result on the **first floor of apartment_0**, including the ability to select the starting point on the map, display the RRT results, and

visualize the robot moving along the planned path.

a. 10 points per object (We will randomly pick 2 out of the 5)
b. 10 points for question and answer.

Report 70%

Your report should include the following content (in English):

1. Implementation (50%)

a. Code

Detailed explanation of your implementation with code screenshots. For example:

- How do you process the 3D pointcloud?
- How do you do the RRT algorithm?

b. Result and Discussion

- Show and discuss the results from the RRT algorithm with different start points and targets.
- Discuss about the robot navigation results.
- Anything you want to discuss.

c. Reference

- Any reference you take

2. Questions (20%)

a. In the RRT algorithm, you can adjust the step size and bias (the number balance between exploration and exploitation). Please explain how the two numbers affect the RRT sampling result. (15%)
b. If you want to apply the indoor navigation pipeline in the real world, what problems may you encounter? (5%)

Bonus 10%

Try to improve the RRT algorithm by comparing and discussing it with the original one.

QA page

The following link is to the QA page for Homework 3. If you have any questions, please post your questions to this Notion page. We will answer them as soon as possible.

Link To Notion Page

Submission

Due date: 2026/4/27

Please directly compress your code files and report (.pdf) into a zip file with this convention {STUDENT_ID}_hw3.zip and submit it to the New E3 System. ex : 123456789_hw3.zip

The file structure should look like:

```
{student_id}_hw3.zip
    ├ README.md (Explain how to run your code)
    ├ report.pdf
    ├ src
        ├ main.py
        └ (other python files, .py format, if you have)
```

- **Wrong submission format leads to -10 points.**
- **Late submission leads to -20 points per day.**
- **Plagiarism is strictly prohibited, including assignments written on GitHub. If we find any instances, it will result in a zero score without any room for objection.**
- **Reports not written in English will receive a score of zero.**

References

[1] RRT